

MARYAM SAIF

AI & Machine Learning Engineer | Data Scientist | Python Developer

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🐙 <https://github.com/saifmaryam> 🌐 <https://saifmaryam.github.io/MaryamSaif.github.io/>

PROFESSIONAL SUMMARY

Highly motivated MS Computer Science graduate specializing in Data Science with strong expertise in Python development, Deep Learning, Explainable AI (XAI), and Medical Image Processing. Demonstrated ability to analyze complex data, build intelligent models, and develop efficient solutions. Research experience in Tri-Modal Malware Detection using CNN, Bi-LSTM, and TabNet with XAI techniques (SHAP, LIME, Grad-CAM).

TECHNICAL SKILLS

Programming Languages:	Python, C++, C#, Java (Basics), HTML, CSS
AI / ML / Deep Learning:	NumPy, Pandas, Scikit-Learn, TensorFlow, PyTorch, OpenCV, MATLAB
Explainable AI (XAI):	SHAP, LIME, Grad-CAM, GNN Explainers, Model Interpretability
Core AI Skills:	Data Preprocessing, Model Training, Transfer Learning, Image Processing, Feature Engineering
Deployment	Hugging Face Spaces, FastAPI, Gradio, Streamlit, REST APIs
Version Control	Git, GitHub repositories, branching, deployment
Databases & Tools:	MySQL, SQLite, MS Office, Android Studio, VS Code, Jupyter Notebook, Google Colab, PyCharm
Soft Skills:	Research Writing, Problem Solving, Analytical Thinking, Team Collaboration

EDUCATION

MS Computer Science (Data Science) — CGPA: 3.25

University of Gujrat, Gujrat, Pakistan 2024 – 2026

BS Computer Science — CGPA: 3.39

University of Sialkot, Sialkot, Pakistan 2020 – 2024

Intermediate (ICS)

Punjab Group of Colleges Daska, Pakistan 2018 – 2020

Matriculation

Govt Girls Islamia High School Daska, Pakistan 2016 – 2018

PROJECTS

Chest X-Ray Disease Detection with Explainable AI (XAI)

PyTorch | OpenCV | Grad-CAM | ResNet18 | FastAPI | Gradio | Hugging Face *Healthcare AI | ✓ Live Deployed*

- Built Deep Learning model detecting Pneumonia/COVID-19 from chest X-rays using ResNet18 Transfer Learning 92%+ accuracy
- Implemented Grad-CAM heatmaps to visualize model focus areas — making AI transparent and interpretable for clinical use
- Deployed as full-stack web application on Hugging Face Spaces with FastAPI backend and animated Gradio frontend
- Dataset: Kaggle Chest X-Ray dataset (5,863 images) — Normal and Pneumonia classes

🔗 **Live Demo:** <https://huggingface.co/spaces/maryam-cheema-ai/PneumoAI>

🔗 **GitHub:** <https://github.com/saifmaryam/PneumoAI>

Food Donation Mobile Application

Android Studio | Java | Firebase | Mobile Development | UI/UX *Term Project*

- Developed mobile application connecting food donors with recipients to minimize food wastage
- Designed user-friendly interface for easy food listing, search, and request management
- Implemented core functionalities: food posting, real-time updates, and user authentication via Firebase
- Promoted social responsibility and community engagement through technology

Customer Churn Prediction with Multi-Model AI Dashboard

Python | Scikit-learn | FastAPI | HTML/CSS/JS | GitHub Pages

Telecom AI | ✓Live Deployed

- Built end-to-end ML pipeline training 3 models (Random Forest 90.3%, Gradient Boosting 91.8%, Logistic Regression 78.1%) on 7,043 telecom customer records with 15 behavioral features
- Engineered real-time REST API using FastAPI with endpoints for single & batch predictions, returning churn probability, risk level (High/Medium/Low), and model accuracy
- Designed interactive dark-themed dashboard (HTML/CSS/JS) with animated probability bars, live model switcher, and instant churn risk visualization zero frameworks used
- Deployed full-stack application with decoupled architecture FastAPI REST backend containerized via Docker on cloud infrastructure and responsive frontend hosted on GitHub Pages fully live at zero cost with auto-redeploy on every Git push
- Dataset: Telecom customer dataset features include contract type, tenure, internet service, monthly charges, and payment method

🌐 **Live Demo:** <https://saifmaryam.github.io/Customer-Churn-Prediction/>

🐙 **GitHub Code:** <https://github.com/saifmaryam/Customer-Churn-Prediction>

Cursor Control System via Hand Gesture Recognition

Python | OpenCV | MediaPipe | Image Processing | Real-time ML

Final Year Project | BS CS

- Developed real-time cursor control system using hand gesture recognition to control mouse movements
- Enabled control of Media Player and PowerPoint slides using specific hand gestures
- Utilized image processing techniques and Python for gesture detection and model training
- Achieved low-latency real-time performance suitable for practical use

MindBot AI — Intelligent Chatbot Platform

React.js | Node.js | MongoDB | REST API | JWT Auth

Full-Stack AI App | In Development

- Developing a full-stack AI chatbot platform featuring complete user authentication (signup, login, JWT-based session management) with a personalized dashboard for each user
- Architecting a decoupled system with a React.js frontend, Node.js/Express REST API backend, and MongoDB database for persistent chat history and user data management
- Integrating AI language models to support multiple conversational personas with real-time response streaming and context-aware dialogue management
- Implementing a tiered access system (Free and Pro plans) with usage tracking, designed for cloud deployment with zero-cost hosting infrastructure

Medical AI Chatbot for Symptom Analysis

Python | HuggingFace Transformers | NLP | Streamlit | Medical NLP | Healthcare AI NLP

NLP | Healthcare AI

- Built AI-powered chatbot analyzing 18+ medical conditions with real-time confidence scores using hybrid NLP and keyword matching engine deployed live on HuggingFace Spaces
- Integrated HuggingFace pre-trained models (biomedical NER, zero-shot classification via BART-MNLI, severity detection via DistilBERT) for symptom extraction and condition mapping
- Designed Streamlit-based web with 4 tabs (Symptom Analysis, Chat, History, Conditions DB) featuring ICD codes, urgency badges, red flags, home remedies, and emergency detection
- Includes medical disclaimer, professional consultation recommendations, and Pakistan emergency contacts (115/1122); fully deployed on HuggingFace Spaces

🌐 **Live Demo:** <https://huggingface.co/spaces/maryam-cheema-ai/medical-ai-chatbot>

🐙 **GitHub Code:** <https://github.com/saifmaryam/MedicalAI>

RESEARCH EXPERIENCE

Tri-Modal Malware Detection

Independent Research — University of Gujrat

2025 – Present

- Performed multimodal malware detection using CNN for image features, Bi-LSTM for dynamic behavioral sequences, and TabNet for static analysis features
- Applied GMU-based multimodal fusion to integrate heterogeneous feature representations for accurate malware classification
- Currently implementing Explainable AI (SHAP, LIME, Grad-CAM, GNN Explainers) to make model decisions transparent and interpretable

LANGUAGES

English — Professional Working Proficiency

Urdu — Native / Full Professional Proficiency